STM32U585 Prosthetic Knee Controller

CubeMX Configuration Reference

Contents

1 Project Setup

Setting	**Value**
Board	B-U585I-IOT02A (or custom STM32U585AI)
MCU	STM32U585AIIx
Project Name	ProstheticKnee_Controller
Toolchain	STM32CubeIDE
HAL Library	STM32Cube_FW_U5
CMSIS	CMSIS-V2
Heap Size	0x4000 (16 KB)
Stack Size	0x800 (2 KB)

2 Clock Configuration (RCC)

Setting	Value
HSE	Crystal/Ceramic Resonator
PLL Source	HSE
SYSCLK	160 MHz
AHB Prescaler	1
APB1 Prescaler	1
APB2 Prescaler	1
Low Speed Clock	LSE (32.768 kHz)

Note: Configure PLL1 from HSE and set SYSCLK to 160 MHz.

3 GPIO Pin Assignments

3.1 LED and Status Outputs

Label	Pin	Mode	Pull	Speed	User Label
Heartbeat LED	PE2 (or PA5)	Output PP	No Pull	Low	LED_HEARTBEAT
Fault LED	PE3	Output PP	No Pull	Low	LED_FAULT
Motor A Enable	PB0	Output PP	No Pull	High	MOTOR_A_EN
Motor B Enable	PB1	Output PP	No Pull	High	MOTOR_B_EN

3.2 Motor Direction Pins

Label	Pin	Mode	User Label
Motor A Direction	PB4	Output PP	MOTOR_A_DIR
Motor B Direction	PB5	Output PP	MOTOR_B_DIR

3.3 Safety Input

Label	Pin	Mode	User Label
Emergency Stop	PC13	Input Pull-Up	ESTOP_INPUT

4 ADC1 Configuration

4.1 General Settings

Setting	Value
ADC Instance	ADC1
Mode	Independent
Clock Prescaler	Sync Clock /4
Resolution	12-bit
Alignment	Right
Continuous Conversion	Enabled
DMA Continuous Requests	Enabled
Oversampling	16x, Right Shift 4
EOC Selection	End of Single Conversion
Scan Direction	Forward
Number of Conversions	3

4.2 ADC Channels

Rank	Channel	Pin	Signal	Sampling Time
1	CH5	PA0	Load Cell Force	640.5 cycles
2	CH6	PA1	Knee Angle Sensor	640.5 cycles
3	CH7	PA2	Spool Position Sensor	640.5 cycles

4.3 ADC DMA

Setting	Value
DMA Request	ADC1
DMA Channel	GPDMA1 Channel 2
Direction	Peripheral to Memory
Mode	Circular
Peripheral Width	Half Word
Memory Width	Half Word
Memory Increment	Enabled

5 USART1 Bluetooth Telemetry

Setting	Value
Mode	Asynchronous
Baud Rate	115200
Word Length	8 Bits
Stop Bits	1
Parity	None
Flow Control	None
TX Pin	PA9 (AF7)
RX Pin	PA10 (AF7)
DMA TX	Enabled

DMA RX	Enabled

5.1 USART1 DMA Allocation

Channel	Direction	Purpose
GPDMA1 CH0	M2P	USART1 TX
GPDMA1 CH1	P2M	USART1 RX

6 USART2 DMC Communication

Setting	Value
Mode	Asynchronous
Baud Rate	115200
Word Length	8 Bits
Stop Bits	1
Parity	None
Flow Control	None
TX Pin	PD5 (AF7)
RX Pin	PD6 (AF7)

6.1 USART2 DMA Allocation

Channel	Direction	Purpose
GPDMA1 CH3	M2P	USART2 TX
GPDMA1 CH4	P2M	USART2 RX

7 PWM Timer Configuration

7.1 TIM1 Motor A PWM

Setting	Value
Timer	TIM1
Channel	CH1
Pin	PE9 (AF1)
Prescaler	159
ARR	999
CCR1	0
PWM Frequency	1 kHz

7.2 TIM2 Motor B PWM

Setting	Value
Timer	TIM2
Channel	CH1
Pin	PA15 (AF1)

Prescaler	159
ARR	999
CCR1	0
PWM Frequency	1 kHz

8 Independent Watchdog

Setting	Value
IWDG	Enabled
Prescaler	/256
Reload Value	1249
Window Value	4095
Timeout	~10 seconds

9 NVIC Priorities

Interrupt	Priority	Sub Priority	Notes
GPDMA1 CH0	5	0	USART1 TX DMA
GPDMA1 CH1	5	0	USART1 RX DMA
GPDMA1 CH2	5	0	ADC DMA
GPDMA1 CH3	5	0	USART2 TX DMA
GPDMA1 CH4	5	0	USART2 RX DMA
USART1	6	0	Idle Detection
USART2	6	0	Idle Detection
ADC1	7	0	ADC Complete
TIM1	8	0	Optional
EXTI13	3	0	Emergency Stop

10 FreeRTOS Configuration

10.1 Kernel Parameters

Parameter	Value
configTICK_RATE_HZ	1000
configMAX_PRIORITIES	7
configTOTAL_HEAP_SIZE	16384
configUSE_PREEMPTION	1
configUSE_TIMERS	1
configUSE_MUTEXES	1
configCHECK_FOR_STACK_OVERFLOW	2
configMAX_SYSCALL_INTERRUPT_PRIORITY	5

10.2 Tasks

Task Name	Priority	Stack	Period

SafetyTask	Realtime (6)	256	1 ms
MotorControlTask	High (5)	256	1 ms
ADCProcessTask	Above Normal (4)	384	5 ms
CommTask	Above Normal (4)	384	10 ms
BluetoothTask	Normal (3)	512	50 ms
WatchdogTask	Low (2)	128	100 ms

11 Queues

Queue Name	Length	Item Size	Purpose
xQueueBT_RX	10	16	Bluetooth RX
xQueueBT_TX	10	16	Bluetooth TX
xQueueComm_RX	10	16	USART2 RX
xQueueSensorData	5	12	Sensor Data
xQueueFaultCode	5	4	Fault Codes

12 Semaphores and Mutexes

Name	Type	Purpose
xMutexUART1	Mutex	USART1 Protection
xMutexUART2	Mutex	USART2 Protection
xSemADC_Ready	Binary Semaphore	ADC DMA Complete
xSemBT_TxDone	Binary Semaphore	UART1 TX Complete

13 Bluetooth Telemetry Packet

```
[ 0xAA ] [ CMD ] [ LEN ] [ PAYLOAD ] [ CHECKSUM ] [ 0x55 ]
```

13.1 Payload Format

Field	Type	Size	Encoding
Spool Angle	$uint8_t$	1	Direct
Force Reading	$int16_t$	2	Direct
Knee Angle	$int16_t$	2	Direct
Moment Reading	$int16_t$	2	Direct
State ID	$uint8_t$	1	Direct
Battery Voltage	$int8_t$	1	x20 Scale

13.2 Packet Layout

```
Byte 0 : 0xAA
Byte 1 : CMD
Byte 2 : LEN = 0x09
Byte 3 : spool_angle
Byte 4-5: force
Byte 6-7: knee_angle
```

```
Byte 8-9: moment
Byte 10 : state_id
Byte 11 : battery_raw
Byte 12 : checksum
Byte 13 : 0x55
```

Total Packet Size = 14 Bytes

14 Complete Pin Summary

Pin	Peripheral	Function	User Label
PA0	ADC1 CH5	Load Cell	ADC_FORCE
PA1	ADC1 CH6	Knee Angle	ADC_KNEE
PA2	ADC1 CH7	Spool Position	ADC_SPOOL
PA5	GPIO	Heartbeat LED	LED_HEARTBEAT
PA9	USART1 TX	Bluetooth TX	BT_TX
PA10	USART1 RX	Bluetooth RX	BT_RX
PB0	GPIO	Motor A Enable	MOTOR_A_EN
PB1	GPIO	Motor B Enable	MOTOR_B_EN
PB4	GPIO	Motor A Direction	MOTOR_A_DIR
PB5	GPIO	Motor B Direction	MOTOR_B_DIR
PC13	EXTI	Emergency Stop	ESTOP
PD5	USART2 TX	DMC TX	DMC_TX
PD6	USART2 RX	DMC RX	DMC_RX
PE2	GPIO	Fault LED	LED_FAULT
PE9	TIM1 CH1	Motor A PWM	PWM_MOTOR_A
PA15	TIM2 CH1	Motor B PWM	PWM_MOTOR_B

15 GPDMA1 Channel Allocation

Channel	Request	Direction	Usage
CH0	USART1_TX	M2P	Bluetooth TX
CH1	USART1_RX	P2M	Bluetooth RX
CH2	ADC1	P2M	Sensor ADC
CH3	USART2_TX	M2P	DMC TX
CH4	USART2_RX	P2M	DMC RX